

S.no	Model Provider	Model Released (Last ~12 Months)	Approx. Release	Primary Focus
1	OpenAI	GPT-5, GPT-5.5	2025–2026	General AI, reasoning, coding, agents
2	Anthropic	Claude Opus 4.7	2026	Advanced reasoning, coding
3	Google DeepMind	Gemini 3.1 Pro	2026	Multimodal, enterprise AI
4	Meta	Muse Spark 1.1	2026	Reasoning and multimodal AI
5	xAI	Grok 4.x	2026	Reasoning, coding, real-time knowledge
6	Mistral AI	Mistral Medium 3.5	2026	Enterprise LLM, coding
7	Alibaba (Qwen)	Qwen3.8-Max	Aug-26	Large-scale multimodal model
8	DeepSeek	DeepSeek V4 Flash / V4 Pro	2026	Cost-efficient reasoning and coding
9	Moonshot AI	Kimi K3	2026	Coding, long-context reasoning
10	Zhipu AI	GLM-5.x	2026	General-purpose multilingual AI

Model Provider	Model	Release Date	Knowledge Cutoff	Cost of Tokens (Input / Output per 1M)	Context Window	Type	When to Use
OpenAI	GPT-5.6 Sol	2026	Not Publicly Disclosed	\$5 / \$30	1M+ tokens	Flagship Reasoning	Advanced reasoning, research, enterprise AI agents, complex coding
OpenAI	GPT-5.6 Terra	2026	Not Publicly Disclosed	\$2.50 / \$15	1M+ tokens	Balanced Reasoning	General-purpose enterprise applications balancing performance and cost
OpenAI	GPT-5.6 Luna	2026	Not Publicly Disclosed	\$1 / \$6	1M+ tokens	Cost-Optimized Reasoning	High-volume chatbots, QA automation, document processing
OpenAI	GPT-5.5	Apr-26	Dec-25	\$5 / \$30	1M+ tokens	Flagship Reasoning	AI agents, advanced coding, research, enterprise automation
OpenAI	GPT-5.5 Pro	Apr-26	Dec-25	\$30 / \$180	1M+ tokens	Premium Reasoning	Scientific research, financial analysis, legal AI, mission-critical systems

OpenAI	GPT-5.4	Early 2026	Not Publicly Disclosed	\$2.50 / \$15	1M+ tokens	General Reasoning	Business applications, workflow automation, document intelligence
OpenAI	GPT-5.4 Pro	Early 2026	Not Publicly Disclosed	\$30 / \$180	1M+ tokens	Premium Reasoning	Enterprise decision support, complex software engineering
OpenAI	GPT-5.4 Mini	Early 2026	Not Publicly Disclosed	\$0.75 / \$4.50	400K tokens	Lightweight Reasoning	Customer support, summarization, RAG applications
OpenAI	GPT-5.4 Nano	Early 2026	Not Publicly Disclosed	\$0.20 / \$1.25	400K tokens	Ultra-Light Model	Mobile apps, edge AI, IoT, high-volume inference
OpenAI	GPT-5.3 Codex	2026	Not Publicly Disclosed	\$1.75 / \$14	400K tokens	Coding Model	Code generation, debugging, unit testing, DevOps
OpenAI	GPT-4.1	Apr-25	Jun-24	\$2 / \$8	1,047,576 tokens	General-Purpose LLM	Business AI, document understanding, chatbots

OpenAI	GPT-4.1 Mini	Apr-25	Jun-24	\$0.40 / \$1.60	1,047,576 tokens	Efficient LLM	Affordable chatbots, automation, knowledge assistants
OpenAI	GPT-4.1 Nano	Apr-25	Jun-24	\$0.10 / \$0.40	1,047,576 tokens	Small LLM	Low-latency AI, embedded assistants, lightweight apps
OpenAI	GPT-Live-1	2026	Not Publicly Disclosed	ChatGPT Only	N/A	Real-Time Voice Model	Natural voice conversations in ChatGPT
OpenAI	gpt-audio-1.5	2026	Not Publicly Disclosed	Text: \$2.50 / \$10	128K tokens	Audio Multimodal	Speech recognition, voice assistants, call-center AI
				Audio: \$32 / \$64			
OpenAI	gpt-audio-mini	2026	Not Publicly Disclosed	Text: \$0.60 / \$2.40	128K tokens	Budget Audio Model	Low-cost transcription and voice bots
				Audio: \$10 / \$20			
OpenAI	gpt-oss-120B	2026	Not Publicly Disclosed	Varies by Provider	131,072 tokens	Open-Weight LLM	Self-hosted enterprise AI, fine-tuning, private deployments

OpenAI	gpt-oss-20B	2026	Not Publicly Disclosed	Varies by Provider	131,072 tokens	Open-Weight LLM	Lightweight on-premise AI applications
OpenAI	GPT Image 2	2026	Not Applicable	Text: \$5	N/A	Image Generation	High-quality image creation, editing, marketing, creative design
				Images: \$8 / \$30			
OpenAI	GPT Image 1.5	2026	Not Applicable	Text: \$5 / \$10	N/A	Image Generation	Cost-effective image generation and editing
				Images: \$8 / \$32			

<https://medium.com/@ashfaqbs/transformers-in-large-language-models-llms-40309d749b47>

<https://www.truefoundry.com/blog/transformer-architecture>

1. What are LLMs?

Large Language Models (LLMs) are computer programs trained on a huge amount of text to generate or understand human-like language.

- They predict the **next word (token)** in a sentence based on the context.
- “Large” means they have **billions of parameters (weights)**.

What are weights?

- Think of weights as the **knobs in a music equalizer** that adjust sound. In an LLM, weights adjust how strongly one word relates to another.
- They are numbers inside the model that get tuned during training so the model can make accurate predictions.

Analogy: An LLM is like a **super-smart autocomplete engine** — but one that can write essays, code, or answer questions.

2. Why are LLMs called “Transformers”?

They’re based on the **Transformer architecture**, introduced in 2017 (*“Attention is All You Need”*).

The name comes from the way the architecture **transforms** input text representations into output text.

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Analogy: Like a factory line where raw material (words) is gradually **transformed** into a finished product (meaningful sentences).

3. Components of a Transformer

(a) Input Embeddings

- Words are turned into **vectors** (lists of numbers) so the model can process them.
- **Who helps?** Weights in the embedding layer map each word to its vector space.
- **Analogy:** Each word is given GPS coordinates in a “map of meaning.”

(b) Positional Encoding

- Transformers don’t naturally know the **order of words**.
- Positional encoding adds order information.
- **Who helps?** Fixed or learned patterns (like sine waves or trainable vectors) added to embeddings.
- **Analogy:** Like numbering lines in a script so actors know which comes first.

(c) Self-Attention

- Core mechanism. It lets the model decide which words are most relevant to each other.
- Uses **Queries, Keys, and Values (Q, K, V)**:
- Query = the word asking for context.
- Key = other words offering context.
- Value = the meaning retrieved.

- **Who helps?** Learned weight matrices turn embeddings into Q, K, and V, and a softmax function distributes attention.
- **Analogy:** When hearing “bank,” your brain checks nearby words like “river” or “money” to figure out which meaning is intended.

(d) Multi-Head Attention

- The model uses multiple “attention heads” to look at relationships from different angles.
- **Who helps?** Each head has its own set of weights, and a combining layer merges them into one output.
- **Analogy:** Like having several detectives, each focusing on different clues (grammar, meaning, topic).

(e) Feed-Forward Networks (FFN)

- Small neural networks inside each layer that refine information.
- **Who helps?** Linear layers (with weights) and activation functions like ReLU.
- **Analogy:** Like refining raw metal into polished tools.

(f) Residual Connections & Layer Normalization

- Keep information stable and prevent “forgetting.”
- **Who helps?** Skip connections pass information forward, and normalization keeps numbers stable.
- **Analogy:** Like saving drafts of an essay at each step so you don’t lose progress.

(g) Decoder (for generation)

- Used in text generation tasks (like GPT). It predicts the next word step by step.
- **Who helps?** Attention layers plus masking (so the model doesn’t peek at future words).
- **Analogy:** Like a storyteller who reads notes and continues the story.